

Question

X is a basic carbonate of metal **M**, possibly containing water of crystallization. Two samples of equal mass were taken and treated separately: one was thoroughly treated in hydrogen, collecting 2.16 g of water, and the remaining residue was the element of metal **M**, weighing 10.36 g; the other was treated in oxygen, and after complete reaction, 0.448 L of carbon dioxide was obtained (101.325 kPa, 0°C), and the remaining residue was the highest oxide of **M**. Then the metal **M** and the carbon:oxygen ratio in **X** are respectively:

A. All other options are incorrect.

B. Cu,1:5

C. Cu,1:8

D. Cu,4:15

E. Mn,1:5

F. Mn,1:8

G. Mn,4:15

H. Zn,1:5

I. Zn,1:8

J. Zn,4:15

K. Pb,1:5

L. Pb,1:8

M. Pb,4:15

N. U,1:5

O. U,1:8

P. U,4:15

Q. La,1:5

R. La,1:8

S. La,4:15

Answer

Correct Answer: **L**

Detailed Explanation

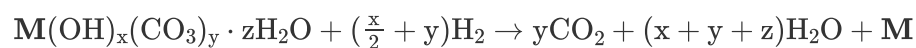
Assuming the chemical formula of **X** is $\text{M}(\text{OH})_x(\text{CO}_3)_y \cdot z\text{H}_2\text{O}$,

CHECKPOINT

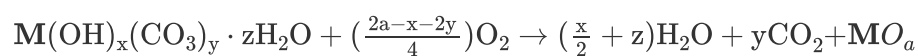
1 PTS

Set up unknowns to represent the chemical formula of **X** as $\text{M}(\text{OH})_x(\text{CO}_3)_y \cdot z\text{H}_2\text{O}$

then the equations for the two reactions are respectively:



and



Thus, we obtain the equation:

$$\frac{x+y+z}{y} = \frac{2.16/18.016}{0.448/22.4}$$

Simplifying, we get $x + z = 5y$

CHECKPOINT

1 PTS

Deduce the relationship between unknowns $x + z = 5y$

From the mass of the metal, we can list the second equation:

$$\frac{10.36}{M} = \frac{2.16}{18.016(x+y+z)} = \frac{2.16}{18.016 \cdot (6y)} = \frac{1}{50.04y}$$

Try to solve the equation. When $y = 2/5$, there is a reasonable solution Pb.

CHECKPOINT

1 PTS

Obtain **M** as Pb by solving the indeterminate equation

At this time, $x + z = 5y = 2$

The common valence states of Pb are +2 and +4. Here, it is obviously +2, so from the valence, we can calculate $x = \frac{6}{5}$, thus $z = \frac{4}{5}$

Therefore, **X** should be $\text{Pb}(\text{OH})_{\frac{6}{5}}(\text{CO}_3)_{\frac{2}{5}} \cdot \frac{4}{5}\text{H}_2\text{O}$.

CHECKPOINT

1 PTS

The chemical formula of **X** is $\text{Pb}(\text{OH})_{\frac{6}{5}}(\text{CO}_3)_{\frac{2}{5}} \cdot \frac{4}{5}\text{H}_2\text{O}$

Therefore, choose option L